

Spreadsheet Applications – Solved Slips

Macro Steps and Output

1. Open Microsoft Excel.
2. Click File → Options.
3. Select Customize Ribbon.
4. Tick the Developer option and click OK.
5. Open Developer tab.
6. Click Visual Basic or press Alt + F11.
7. Click Insert → Module.
8. Type the macro code:

```
Sub CheckValue()
```

```
If Range("A1").Value > 70 Then
```

```
MsgBox "Greater than 70"
```

```
End If
```

```
End Sub
```

9. Press Ctrl + S to save file.
 10. Save as Excel Macro-Enabled Workbook (.xlsm).
 11. Enter any number in cell A1.
- Example: 85
12. Go to Developer → Macros.
 13. Select CheckValue macro.
 14. Click Run.

Output :

If value in A1 is greater than 70, message box displays:

“Greater than 70”

Slip 1

Q1. Apply VLOOKUP function : Search Grade for score 89

```
=VLOOKUP(89,C2:D7,2,TRUE)
```

Answer → Grade = B

Q2. Create Combo Chart

Steps:

1. Select data
2. Insert → Combo Chart
3. Sales = Column Chart
4. Profit = Line Chart

Q3. Apply IF(), NOT(), AND(), OR()

=IF(C2>=50,"Pass","Fail")

=NOT(C2<50)

=AND(C2>=50,D2>=60)

=OR(C2>=90,D2>=90)

Slip 2

Q1. Apply CONCAT() and LEN()

=CONCAT(B2," ",C2)

=LEN(B2)

Q2. Apply Array Formula using IF()

=SUM(IF(B2:B7>40,B2:B7,0))

Press Ctrl + Shift + Enter

Answer → 209

Q3. Create Histogram

Steps:

1. Select marks data
 2. Insert → Histogram Chart
-

Slip 3

Q1. Apply LEFT() and RIGHT()

=LEFT(B2,3)

=RIGHT(C2,3)

Q2. Draw Histogram

Steps:

1. Select attendance data
2. Insert → Histogram

Q3. Apply MID() Array Formula

=MID(A1:A5,1,3)

Press Ctrl + Shift + Enter

Slip 4

Q1. Apply MID() and TRIM()

=MID(B2,2,3)

=TRIM(B2)

Q2. Calculate Cube of Expenses

=B2^3

OR

=POWER(B2,3)

Q3. Create Pivot Table

Steps:

1. Insert → Pivot Table

2. Subject → Rows

3. Marks → Values

4. Set Value Field = MAX

Slip 5

Q1. Apply VLOOKUP : Search Grade for score 56

=VLOOKUP(56,C2:D7,2,TRUE)

Answer → Grade = E

Q2. Create Histogram Chart

Steps:

1. Select marks data

2. Insert → Histogram

Q3. Student Academic Details

=SUM(E2:I2)

=(J2/500)*100

=IF(K2>=40,"Pass","Fail")

Slip 6

Q1. Apply VLOOKUP : Search Grade for score 77

=VLOOKUP(77,C2:D7,2,TRUE)

Answer → Grade = C

Q2. Create Waterfall Chart

Steps:

1. Select marks data
2. Insert → Waterfall Chart

Q3. Create Pivot Table

Steps:

1. Insert Pivot Table
 2. Subject → Rows
 3. Marks → Values → MAX
-

Slip 7

Q1. Apply Formatting to Table

Steps:

1. Bold headings
2. Center align data
3. Font size = 14
4. Apply borders
5. Change heading color

Q2. Apply Mathematical Functions

```
=SUM(B2:F2)  
=PRODUCT(B2:F2)  
=SQRT(G2)  
=POWER(B2,2)
```

Q3. Calculate Cube of Expenses

```
=B2^3
```

Slip 9

Q1. Apply IF() and IFS()

=IF(C2>=50,"Pass","Fail")

=IFS(C2>=90,"A",C2>=75,"B",C2>=60,"C",TRUE,"Fail")

Q2. Apply Mathematical Functions

=SUM(B2:F2)

=PRODUCT(B2:F2)

=ROUND(G2,2)

=POWER(B2,2)

Q3. Employee Performance Status

=IF(J2>=60,"Excellent","Needs Improvement")

Slip 12

Q1. Apply Array Formula

=SUM(A2:A6*B2:B6)

Press Ctrl + Shift + Enter

Answer → 329

Q2. Sort and Filter Data

Steps:

1. Data → Sort → Descending
2. Apply Filter → Attendance below 75%

Q3. Apply IF(), AND(), OR(), NOT()

=IF(C2>=50,"Pass","Fail")

=AND(C2>=50,D2>=60)

=OR(C2>=90,D2>=90)

=NOT(C2<50)

Slip 14

Q1. Apply IF() and AND()

=IF(C2>=50,"Pass","Fail")

=AND(C2>=50,D2>=60)

Q2. Create Combo Chart

Steps:

1. Insert → Combo Chart
2. Sales = Column
3. Profit = Line

Q3. Apply UNIQUE and SORT

=SORT(UNIQUE(A2:A8))

Slip 16

Q1. Apply OR() and NOT()

=OR(C2>=90,D2>=90)

=NOT(C2<50)

Q2. Create Combo Chart

Steps:

1. Insert → Combo Chart
2. Income = Column
3. Growth = Line

Q3. Create Pivot Table

Steps:

1. Department → Rows
 2. Performance Score → Values → MAX
-

Slip 18

Q1. Apply IFS() and OR()

```
=IFS(C2>=90,"A",C2>=75,"B",TRUE,"Fail")  
=OR(C2>=90,D2>=90)
```

Q2. Apply FILTER Function

```
=FILTER(A2:B10,B2:B10="Science")
```

Q3. Multiply Two Columns Using Array Formula

```
=A1:A5*B1:B5
```

Press Ctrl + Shift + Enter

Slip 19

Q1. Write Macro to check value >70

```
Sub CheckValue()  
If Range("A1").Value > 70 Then  
MsgBox "Greater than 70"  
Else  
MsgBox "Less than or equal to 70"  
End If  
End Sub
```

Q2. Create Gantt Chart

Steps:

1. Insert → Stacked Bar Chart
2. Hide Start Date series
3. Format chart

Q3. Apply LEN(), MID(), LEFT(), RIGHT()

```
=LEN(B2)  
=MID(B2,2,3)  
=LEFT(B2,3)  
=RIGHT(C2,3)
```

Slip 21

Q1. Write Macro to check value >70

```
Sub Check70()  
If Range("A1").Value > 70 Then  
MsgBox "Greater than 70"  
Else  
MsgBox "Not Greater than 70"  
End If  
End Sub
```

Q2. Create Histogram Chart

Steps:

1. Select data
2. Insert → Histogram

Q3. Apply CONCAT(), MID(), TRIM(), LEN()

```
=CONCAT(B2,C2)  
=MID(B2,2,3)  
=TRIM(B2)  
=LEN(B2)
```

Slip 29

Q1. Apply Formatting to Table

Steps:

1. Bold headings
2. Center align
3. Font size = 14
4. Apply borders
5. Change heading color

Q2. Write Grade System Macro

```
Sub GradeSystem()  
Dim m As Integer  
m = Range("A1").Value
```

```
If m >= 90 Then  
MsgBox "Grade: A+"  
ElseIf m >= 75 Then  
MsgBox "Grade: A"  
ElseIf m >= 60 Then  
MsgBox "Grade: B"  
ElseIf m >= 40 Then  
MsgBox "Grade: C"  
Else  
MsgBox "Fail"  
End If  
End Sub
```

Q3. Apply Array Formula

```
=SUM(IF(A1:A5>10,A1:A5,0))
```

Press Ctrl + Shift + Enter